IBM Project 1 - Validation and AI-use Record

Prepared: October 14, 2026, America/Indiana/Indianapolis
Model repository baseline: `0e3981bd1709ad0c35ef3eff8583c1d3afd0ca02`
Valuation cutoff: September 1, 2026, after close
Decision: Watch / Defer IBM common stock. This is a validation record, not a replacement investment decision.

1. Final model and execution environment

The final connected model uses USD millions except per-share data. It was independently executed for this record with the repository's `python3` command on macOS (Darwin 24.6.0 arm64), Python 3.14.7. The project requires Python 3.10+ and the standard library only.

Current calculated output	Result
IBM reference close	$231.40
Low / base / high DCF	$145.1346 / $182.2876 / $224.2811 per share
Base WACC / operating EV	7.0708% / $220,096.1m
Terminal PV share	78.0637%
September 1 cash / common equity	$9,280.5m / $173,665.4m
FY2026 company-style FCF proxy / remaining gap	$10,918.2m / $(4,781.8)m
Reverse-DCF terminal growth	3.8150%, holding 15% terminal incremental ROIC constant

2. Sources and SBC treatment

Item	Source and treatment
H1 2026	IBM Q2 2026 Form 10-Q, filed July 23, 2026: June 30 balance sheet, H1 results, $7,766m CFO, and $1,004m SBC.
FY2025	IBM FY2025 Form 10-K, filed February 24, 2026: FY2025 history and $1,715m SBC; H1 2025 $842m implies H2 2025 $873m.
SBC schedule	H2 2026 $938.5m: midpoint of reported H1 2026 $1,004m and historical H2 2025 $873m. FY2027-30 $1,877m each year, provisional and held flat.
Management FCF definition	$7,766m GAAP CFO - $2,264m Financing-receivable adjustment - $743m net capital investment = $4,759m, rounding to reported $4.760bn. Consolidated CFO, management FCF, and operating FCFF are distinct.

Forecast SBC is assumed embedded in forecast operating-cost ratios. The model adds it once to consolidated CFO as a noncash adjustment and credits it once to APIC/equity. It is not a cash share issuance and is not added again to operating FCFF. The repair raises September 1 cash by $321.3m, or about $0.34 per share, but leaves operating FCFF and operating EV unchanged. Future dilution remains unresolved.

3. Validation register: independently executed October 14, 2026

Check	Expected result	Actual result and evidence
Automated suite	30 passing tests	**PASS** - 30 passed, 0 failed; `python3 -m unittest discover -s tests -v`
Published Week 3 known-answer case	All 12 outputs reconcile; 11% WACC decreases value and 10% restores baseline	**PASS** - all 12 within tolerance; $23.4140/share at 11%; $27.4974/share restored at 10%. See `docs/week3-known-answer-reconciliation.md`.
Three-statement articulation	Balance-sheet and cash-flow gaps equal zero	**PASS** - FY2026-30 articulate within floating-point rounding.
Historical reconciliation	Opening, revenue, pretax, and H1 FCF reconcile	**PASS** - 0 / 0 / 0.0 bridge gaps; $4,759m H1 FCF rounds to $4,760m.
Enterprise-to-equity bridge	$182.2876 base value	**PASS** - $220,096.1m EV, $2,740.5m excess cash/securities, $173,665.4m common equity, 952.7m shares.
WACC and terminal direction	Value falls as WACC rises and rises with growth/ROIC	**PASS** - monotonicity and terminal-reinvestment tests passed.
Peer arithmetic	Separate peer bridge consistency	**PASS** - peer implied-value reconciliation test passed.
Scenarios	Low < base < high	**PASS** - $145.13 < $182.29 < $224.28.
September 1 stub	121/184 of H2 FCFF, from September 2	**PASS** - $4,791.1m residual FCFF.
Deliberate failure and recovery	$1m FY2027 cash break fails safely, normal model recovers	**PASS** - expected `$-1.0000m` balance-sheet failure; normal suite remained articulated.
SBC controls	One CFO add-back/APIC credit; no H1/H2 double count; no FCFF add-back	**PASS** - four dedicated SBC tests passed.

`python3 model/ibm_phase3_model.py --write-output --demo-failure` printed the expected safe failure and regenerated output. A local workbench API request on port 8770 returned `statement_articulation: PASS` and base value `$182.287630`.

> **Historical versus task-run evidence.** This task did not repeat a fresh GitHub download. The October 14 fresh-download/launcher evidence in `docs/cold-run-log.md` is student-observed pre-SBC evidence: the launcher selected port 8768, the base was $181.95, and 24 tests were recorded. It is not represented as a final-model task-run result.

Complete test list: all returned `ok`: `test_all_years_articulate`, `test_annual_depreciation_after_2026`, `test_bounds_fail`, `test_december_cash_uses_h2_only`, `test_december_equity_uses_h2_only`, `test_fy26_investment_h1_plus_h2`, `test_fy26_net_income_h1_plus_h2`, `test_fy26_revenue_h1_plus_h2`, `test_historical_base`, `test_no_unallocated_corporate_deduction`, `test_one_million_cash_break_fails`, `test_operating_fcff_excludes_financing`, `test_reported_financing_equity`, `test_residual_fcff_not_full_year_prorated`, `test_residual_fcff_uses_h2`, `test_sbc_does_not_change_operating_fcff`, `test_sbc_historical_anchor_and_h2_schedule`, `test_sbc_is_noncash_cfo_addback_with_matching_apic`, `test_cases_reverse_dcf_and_named_scenario_order`, `test_dates_and_correct_stub_convention`, `test_peer_and_implied_value_reconciliations`, `test_prevaluation_rollforward_changes_bridge_not_operating_fcff`, `test_sbc_changes_cash_bridge_not_operating_ev_or_stub`, `test_sep1_bridge_rollforward_and_dividend_once`, `test_terminal_monotonicities`, `test_terminal_reinvestment_and_boundaries`, `test_wacc_and_growth_value_monotonicity`, `test_wacc_uses_corporate_debt_and_marginal_tax`, `test_published_twelve_outputs`, and `test_wacc_sensitivity_decreases_value_and_baseline_restores`.

4. Historical Locked Changed-Input Record

The original locked text files and four screenshots remain unchanged in `docs/`. They document a pre-SBC experiment, not the current $182.29 baseline.

Historical field	Preserved record
Prediction / execution	October 10, 2026, 12:39pm, AI off / 12:53pm.
Input	Software revenue growth: 0pp to +2pp for FY2027-30.
Prediction	Revenue, operating profit, FCFF, and intrinsic value would rise; Watch / Defer would not change.
Observed result	$181.95 to $190.31/share; FY2030 EBIT $15,279.2m; FCFF $14,152.8m; WACC 7.07%.
Decision effect	Prediction direction matched. Value increased $8.36/share (4.6%) but remained below $231.40; Watch / Defer remained.
Provenance	Original predecessor repository prediction commit `7359ba2` is reported but unavailable here. Archived BEFORE commit `4591725`, timestamped 18:21:56 -04:00, is a later preservation event, not the original 12:39pm prediction time. Results/screenshots commit: `2f2384a`.

5. AI-use record and limitations

OpenAI Codex assisted under student direction with code inspection, test execution, documentation, and HTML formatting. It did not obtain course answers, submit work, trade securities, or access a proprietary terminal. Exact model/version metadata beyond the product identity was not supplied and is not claimed. The student remains responsible for source verification, assumptions, judgment, and submission.

- The September 1 timing correction from a 122/184 to a 121/184 FCFF stub was AI-assisted, independently checked, and enforced by tests.
- The student approved the October 14 SBC review. Dedicated tests verify the one-time CFO/APIC treatment and no operating-FCFF double count.
- Limitations remain: $4.782bn management-FCF gap, future dilution/share-count uncertainty, qualified beta and issuer-specific borrowing-cost evidence, 78.06% terminal-value dependence, pension convention, and IBM Financing book-equity sidecar.
- Escalate or re-underwrite if reported growth, margins, cash conversion, debt, dilution, or WACC evidence materially contradicts the model. The current action remains Watch / Defer.

6. Known-answer status

The published FIN 43900 Week 3 training case and answer table were located in the course repository. A standalone conventional-DCF implementation calculated all 12 outputs from published inputs; every comparison passed the documented tolerance. At 11% WACC, value was $23.4140/share, below the restored 10% baseline of $27.4974/share. See `docs/week3-known-answer-reconciliation.md` and `docs/known-answer-blocker.md` for sources, comparisons, commands, and scope.

This resolves the published Week 3 known-answer check. It does not make the training case an end-to-end test of IBM-specific timing, ROIC-terminal, Financing, or forecast logic. This PDF was generated from the corresponding HTML validation record. The Markdown source provides additional supporting detail.